ModelIt!

Teacher's Guide

DNA Decoded: Why You Are (Mostly) Unique

The Blueprint of Life and What Makes You, You

Quick Reference

Grade Level	Grade 07L2
NGSS Standard	MS-LS3-1, MS-LS3-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-LS3-1, MS-LS3-2

Develop and use a model to describe why structural changes to genes (mutations) located on chromosomes may affect proteins and may result in harmful, beneficial, or neutral effects to the structure and function of the organism. Develop and use a model to describe why asexual reproduction results in offspring with identical genetic information and sexual reproduction results in offspring with genetic variation.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop and use a model to describe how genetic variation arises through sexual reproduction and how mutations affect organisms.
Disciplinary Core Idea	**LS3.A Inheritance of Traits / LS3.B Variation of Traits** — Genes are located on chromosomes. Variations of inherited traits can be explained by sexual reproduction and mutations.
Crosscutting Concept	**Cause and Effect** — Students identify how parent gene variation and environmental conditions cause differences in trait expression and survival outcomes.

CAST Testing Alignment

MULTIPLE CHOICE:

A farmer plants a field with genetically identical crop plants. A new fungal disease arrives. What is the most likely outcome and why?

CONSTRUCTED RESPONSE:

Using your genetics model, explain why sexual reproduction produces offspring with more genetic variation than asexual reproduction. Include how this variation affects a population's ability to survive environmental changes.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

DNA is the molecule that carries the instructions for building and maintaining every living organism. Understanding how genes are inherited, expressed, and varied is fundamental to explaining why organisms look and function the way they do.

DNA and Genes

DNA (deoxyribonucleic acid) is a double-helix molecule containing the instructions for building proteins. Genes are specific segments of DNA that code for particular traits. Humans have approximately 20,000 genes spread across 23 pairs of chromosomes.

Sexual vs. Asexual Reproduction

In sexual reproduction, offspring receive half their DNA from each parent through meiosis, creating unique combinations every time. In asexual reproduction, a single parent produces genetically identical offspring. Sexual reproduction is slower but creates diversity; asexual is faster but creates clones.

Gene Expression and Environment

Not all genes are active at all times. Environmental factors like nutrition, temperature, stress, and sunlight can turn genes on or off through a process called epigenetics. This means identical DNA can produce different results in different environments, which is why identical twins raised apart may develop differently.

Mutations: The Source of New Variation

Mutations are permanent changes in DNA that occur during cell division. Most mutations are neutral (no effect), some are harmful (cause disease), and rarely, some are beneficial (improve survival). Over generations, beneficial mutations can spread through a population through natural selection.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify parent gene variation, environmental conditions, gene expression level, trait variation, and survival advantage as components of the genetic system.

E **Establish Relationships**
Students determine that parent gene variation drives trait diversity, while environmental conditions influence which traits are expressed and which provide survival advantages.

V **Visualize & Model**
Students build a model showing how genetic and environmental inputs interact to produce diverse organisms with varying survival prospects.

E **Evaluate Outcomes**
Students run high variation, environmental shift, and low variation scenarios to observe how genetic diversity affects population resilience.

R **Revise & Extend**
Students add mutation rate or epigenetic factors to explore how new genetic variation arises and how gene expression can change without DNA changes.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"You share 99.9% of your DNA with every person in this room. So what makes you, YOU?"*

Watch For

Show a side-by-side of diverse students. Ask: What makes each person unique?

Slide 2: Learning Objectives

3 min

Say: *"Today we're decoding the blueprint of life to understand why you are mostly unique."*

Watch For

Have students read objectives. Pre-teach 'gene expression' and 'allele.'

Slide 3: Big Question

3 min

Say: *"That 0.1% difference? It contains MILLIONS of variations. Let's explore what they do."*

Watch For

Show a class trait survey (eye color, hair type, etc.). Discuss visible variation.

Slide 4: LEVER Framework

2 min

Say: *"We'll build a model that shows how genes and environment interact to create YOU."*

Watch For

Preview LEVER steps. Connect to prior knowledge about inherited traits.

Slide 5: Activity 1: Components

8 min

Say: *"What inputs determine your traits? What can nature control vs. what responds automatically?"*

Watch For

Guide sorting. Discuss why parent genes and environment are external inputs.

Slide 6: Activity 2: Connections

8 min

Say: *"When parent gene variation is high, what happens to offspring diversity? And how does environment change the picture?"*

Watch For

Help students trace relationships. Discuss positive and negative connections.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's compare a population with high genetic diversity to one with almost none. What happens when the environment shifts?"*

Watch For

Students predict first, then run simulations. Compare survival outcomes.

Slide 8: Discoveries

5 min

Say: *"Why is genetic diversity a species' insurance policy? What did our model reveal?"*

Watch For

Lead discussion about diversity and survival. Connect to real examples.

Slide 9: STEM Challenge

12 min

Say: *"You're now museum designers. Build an exhibit that shows why genetic diversity is essential."*

Watch For

Groups design interactive exhibits using model data.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Parent Gene Variation, Environmental Conditions
- **INTERNAL:** Gene Expression Level, Trait Variation in Offspring, Survival Advantage

Relationships

- **Parent Gene Variation to Trait Variation in Offspring:** POSITIVE (+) — When parents have more diverse alleles, their offspring have a wider range of possible trait combinations, increasing diversity in the population.
- **Environmental Conditions to Gene Expression Level:** POSITIVE/NEGATIVE (+/-) — Environmental factors can turn genes on or off. Favorable conditions may activate growth genes while harsh conditions may activate stress-response genes.
- **Gene Expression Level to Trait Variation in Offspring:** POSITIVE (+) — When more genes are actively expressed, there are more visible and functional trait differences among offspring.
- **Trait Variation in Offspring to Survival Advantage:** POSITIVE/NEGATIVE (+/-) — Whether trait variation helps or hurts depends on the environment. In a changing environment, more variation means a better chance that some individuals have the right traits.
- **Environmental Conditions to Survival Advantage:** POSITIVE (+) — Favorable environmental conditions improve survival for all organisms, regardless of their specific traits.

Key Discoveries

- Sexual reproduction creates unique combinations of parent genes, which is why siblings look different despite sharing parents
- Environmental conditions can turn genes on or off, meaning identical DNA can produce different traits in different environments
- Populations with more genetic variation are better equipped to survive environmental changes
- Asexual reproduction creates identical copies, which is efficient but dangerous if the environment changes

Multiple Choice Post-Assessment Answer Key

1. A farmer plants a field with genetically identical crop plants (produced by asexual reproduction). A new fungal disease arrives and destroys the entire crop. Based on the genetics model, which explanation best accounts for why the entire field was lost?

Correct: B) All plants had identical DNA, so if one lacked resistance to the fungus, none of them had resistance

Correct! With zero genetic variation, all individuals share the same vulnerabilities. If one plant lacks resistance to the fungus, every plant in the genetically identical population lacks it. This is the fundamental danger of low genetic diversity.

2. Two siblings share the same parents but look noticeably different from each other. According to the model, which factor is most responsible for their physical differences?

Correct: B) Sexual reproduction randomly combines different allele combinations from each parent during meiosis, creating unique offspring every time

Correct! During meiosis, each parent contributes a random half of their chromosomes. Crossing over further shuffles genes. With millions of possible allele combinations, each sibling receives a unique genetic hand, which is why they look different despite sharing parents.

3. In the model, when Environmental Conditions change but Parent Gene Variation stays constant, some offspring gain a Survival Advantage while others lose it. Which crosscutting concept best explains this observation?

Correct: B) Cause and Effect: the environmental change causes certain trait combinations to become advantageous or disadvantageous

Correct! The Cause and Effect relationship is central: a change in environmental conditions (cause) shifts which traits provide survival advantages (effect). A trait that was neutral or harmful in one environment may become beneficial in another.

4. A student argues that a mutation that causes sickle-shaped red blood cells is always harmful. Using the model's concept of environmental influence on survival advantage, which response best challenges this claim?

Correct: B) In regions with malaria, carrying one copy of the sickle cell allele provides resistance to the disease, making the same mutation beneficial in that environmental context

Correct! Whether a trait is harmful or beneficial depends on the environment. The sickle cell allele in its full form causes disease, but carrying just one copy provides malaria resistance. In malaria-prone regions, this mutation increases survival advantage, demonstrating that context determines whether a trait helps or harms.

5. In the model, when Parent Gene Variation is set to low (simulating asexual-like reproduction), Trait Variation in Offspring is minimal and Survival Advantage drops sharply when Environmental Conditions change. A student concludes that asexual reproduction is always inferior to sexual reproduction. Why is this conclusion too broad?

Correct: B) Asexual reproduction is highly efficient in stable environments because identical offspring are already well-adapted, but it becomes dangerous when conditions change

Correct! Asexual reproduction is efficient in stable environments because every offspring is already well-adapted (no wasted variation). But this advantage becomes a liability when the environment changes, because the population lacks the diversity needed to adapt. Neither strategy is universally superior.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Traits are only from one parent"	Every trait is influenced by genes from BOTH parents. You receive one allele from your mother and one from your father for each gene. Which allele is expressed depends on dominance patterns and sometimes both alleles contribute (codominance).
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"DNA determines everything"	While DNA provides the blueprint, environmental conditions strongly influence which genes are expressed and how. Nutrition, stress, exercise, and exposure to chemicals can all change gene expression without changing the DNA itself. You are not ONLY your genes.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Mutations are always bad"	Most mutations are neutral and have no effect. Some are harmful, causing diseases. But some mutations are beneficial, providing new traits that help organisms survive. All genetic diversity ultimately comes from mutations. Without them, evolution couldn't occur.